JMP TEST 2 · FRONT

TWO-SAMPLE T · MATCHED PAIRS | ▼ RED TRIANGLE · › THEN CLICK · **GREEN = WRITE IT · FAINT WORD IN A BLANK = WHAT GOES THERE**

SYMBOLS

μ "mew", a **Greek** letter the **TRUE** mean of **everyone** — you never know it
RULE H_0 and H_1 are **always** about μ, **never** about $\bar{x}$
s_p pooled SD — one SD for both groups
α "alpha" the bar you compare p to
MSE Mean Square Error = s_p^2
HSD Tukey cutoff; |diff| > HSD = different
Prob > F the ANOVA p-value

$\bar{x}$ "x-bar" the mean of **your sample** — you measured it, you know it
$\bar{d}$ "d-bar" average of the differences
SE standard error = spread of the estimate
H_0 the "no difference" **tie** — always has an =
RMSE √MSE, the typical error
LSMean least squares mean = a cell average
A*B the interaction term

μ is NOT the letter u write it like a u with a tail hanging down
μ_d true mean difference; H_0: $\mu_d = 0$
df 2-sample n_1+n_2-2 · paired n−1
H_1 "higher"→**>** · "lower"→**<** · "different"→**≠**
k how many groups
Prob > |t| the **two-sided** p-value
t* critical t — needs df

μ_1 μ_2 group 1 and group 2 true means — always say **which is which**
s_1 s_2 the two sample SDs
n_1 n_2 the two sample sizes
F ANOVA test statistic; big F = real effect
q studentized range — Tukey uses it
Prob > t one-sided p, upper tail

THE DECISION RULE — same for all four tests

Compare p to α. That is the whole decision.

`P < A` **REJECT H_0** → "there **IS** sufficient evidence that …"
`P > A` **FAIL TO REJECT H_0** → "there is **NOT** sufficient evidence that …"

Never write "**accept** H_0" and never write "**prove**". You only ever reject or fail to reject.

Comparing the decimals — line them up

`EXAMPLE` α = 0.0**100** · p < 0.**0001** → p is **smaller** → **reject**

More zeros after the point = smaller number. 0.0001 is 1 in 10,000; 0.01 is 1 in 100. A **tiny p is strong evidence against H_0.**

JMP prints `0.0000` or `<.0001` because it ran out of decimals — you write **p < 0.0001**, never p = 0.
Same p can flip the answer: p = 0.032 **rejects** at α=0.05 but **fails to reject** at α=0.01. **Find α in the question first.**

Write it p first

`SAY` Since p *p-value* is **less than** α = *.05 or .01* , we **reject H_0**. There is sufficient evidence that *H_1 in plain words* .

`SAY` Since p *p-value* is **greater than** α = *.05 or .01* , we **fail to reject H_0**. There is not sufficient evidence that *H_1 in plain words* .

One-sided? Name the **direction** ("higher for the new method"), not just "different".

WHICH TEST? — decide this first, every time

Count the variables, then ask who the subjects are

`2 GROUPS` **different** subjects → two-sample t › Fit Y by X
`2 GROUPS` **same** subject twice → Matched Pairs
`1 FACTOR` 3+ groups → one-way ANOVA › Fit Y by X
`2 FACTORS` → two-way ANOVA › Fit Model › Cross

Tie-breaker: could you shuffle one column without losing info? Yes → two-sample. No → matched pairs.

Data layout is OPPOSITE for those two

`FIT Y BY X` **stacked** — 1 response column + 1 group column
`MATCHED PAIRS` **side by side** — 2 response columns

Column won't go in the X box? Right-click header › Column Info › Modeling Type → **Nominal**. Text won't type in? Data Type → **Character**.

T1 · TWO INDEPENDENT MEANS — two-sample t

a Raw data, or just summary numbers?

`RAW ROWS` Analyze › Fit Y by X › Y = response, X = group › OK
`ONLY N, X̄, S` Help › Sample Index › Tool › Calculators › **TestTwoSampleMeans** › t-test › Assume Equal Variances › pick H_1

Fit Y by X cannot run on summary statistics. It needs actual rows. **The calculator gives you NO Brown-Forsythe and NO quantile plot** — argue conditions from the problem text and the SD ratio by hand.

b Turn on everything you need, once

▼ Means/Anova/Pooled t · t Test · Unequal Variances · Normal Quantile Plot

That one click-set answers conditions, test statistic, p-value and CI.

c Variables and parameters

`SAY` The explanatory variable is *what splits the groups* , which is **categorical** with 2 levels. The response is *what you measured + units* , which is **quantitative**.

`SAY` μ_1 = the true mean *what + units* for **all** *group 1* , and μ_2 = the true mean for **all** *group 2* .

`SAY` The point estimate is $\bar{x}_1 - \bar{x}_2$ = *mean 1* − *mean 2* = *answer* .

Parameters are about the whole population, never the numbers you were handed.

d Hypotheses — watch the wording

`TWO-SIDED` "a difference", "differ", "changed" → $H_1: \mu_1 \neq \mu_2$
`ONE-SIDED` "higher", "greater", "more than", "improved" → $H_1: \mu_1 > \mu_2$

`SAY` H_0: $\mu_1 = \mu_2$ H_1: μ_1 *< > or ≠* μ_2

e Conditions — all three, in this order

`SAY` **Independence:** subjects were **randomly assigned** (or randomly sampled) and each appears in only one group, so the two samples are independent.

`SAY` **Normality:** n = *group size* per group. *If n ≥ 30:* the Central Limit Theorem applies, so no plot is needed. *If n < 30:* the Normal Quantile Plot shows the points following their lines with no curve or outliers.

`SAY` **Equal variances:** the SDs are *s1* and *s2* , a ratio of *bigger÷smaller* which is under 2, and Brown-Forsythe gives p = *p-value* > 0.05, so pooling is appropriate.

SD ratio under **2** = variance ratio under **4**. Fails? Don't stop — quote the **t Test** (Welch) box instead of Pooled t. Conditions 1 and 2 are required either way.

f Test statistic and p-value

`EQUAL VAR` **Pooled t Test** box → t Ratio, DF = n_1+n_2-2, Prob > |t|
`UNEQUAL` **t Test** box ("Assuming unequal variances") → decimal DF

`SAY` If H_0 were true, there would be about a *p as a %* chance of getting a difference in sample means this far from 0 just by random chance.

One-sided? Use **Prob > t** or **Prob < t**, never Prob > |t|. Read the header (e.g. *B−A*) to see the subtraction order. Safety check: the right one-sided p is **below 0.5** when the data lean the way H_1 says. Got 0.9997? Take the other row.

g Conclusion

Use **The decision rule** at the top of this page: compare p to α, then write the sentence **in context**, naming the direction.

h Confidence interval

`READ` **Upper CL Dif** and **Lower CL Dif** in the same t-test box

`SAY` We are 95% confident the true difference in mean *what + units* is between *Lower CL* and *Upper CL* .

CI excludes 0 ↔ you rejected H_0. They always agree. If asked "will it contain 0?" before computing → answer from the p-value.

T2 · MATCHED PAIRS — same subject measured twice

a Run it

`PATH` Analyze › **Specialized Modeling** › **Matched Pairs**
`THEN` put **BOTH** columns into **Y, Paired Response** › OK

One box, two columns. Do **not** stack this data.

b Equivalent route, if asked the long way

`MAKE COL` Cols › New Column › Column Properties › Formula → after − before
`THEN` Analyze › Distribution › ▼ › Test Mean › 0

Identical t and p. The difference column is also how you check normality.

c Hypotheses

`SAY` Let μ_d be the true mean difference (*after* minus *before*) for all *subjects* .

`SAY` H_0: $\mu_d = 0$ H_1: μ_d *< > or ≠* 0

d Conditions — only two, and one is different

`SAY` **Independence:** the **pairs** were randomly sampled and one pair does not affect another. The two values *within* a pair are dependent by design, which is the point.

`SAY` **Normality:** the **differences** are approximately normal — the quantile plot of the difference column shows no curve or outliers.

Normality applies to the **differences**, not to either original column. There is **no** equal-variance condition here.

e Read it and conclude

`READ` Mean Difference · t-Ratio · **DF = n−1** · Prob > |t| · Upper95% / Lower95%

5 pairs → **df = 4**, not 8. JMP prints the subtraction order in the label (e.g. *2022-2020*); the second column minus the first flips only the **sign**, never the p-value.

`SAY` Since p = *p-value* < α = *.05* , we reject H_0. There is sufficient evidence of a difference in mean *what + units* between *time 1* and *time 2* .

f Confidence interval for the difference

`READ` **Upper95%** and **Lower95%** in the same box

`SAY` We are 95% confident the true mean difference in *what + units* is between *Lower95%* and *Upper95%* . It **excludes 0**, agreeing with rejecting H_0.

g Why pairing is worth it

Pairing strips out subject-to-subject variation, so t gets bigger. Run wrongly as two independent samples the same data gives a much weaker result. That's the answer if asked why you paired.

WORKED EXAMPLE 1 · SNAP BEANS — two-sample t

Given

`DATA` Old: n=32 x̄=410 s=30 · New: n=32 x̄=449 s=25 · α=0.01
`PATH` Calculators › TestTwoSampleMeans › t-test › Equal Variances › **upper tail** (New = Sample 1)

Answers

`A` Method is **categorical**, 2 levels. Ascorbic acid (mg/kg) is **quantitative**.

`B` μ_1 = true mean ascorbic acid for **all** packages stored the new way; μ_2 = same for the old way.

`C` H_0: $\mu_1 = \mu_2$ H_1: $\mu_1 > \mu_2$ *("higher" = one-sided)*

`D` 64 pkgs **randomly assigned** ✓ · n=32 ≥ 30 so **CLT** covers normality, no plot needed ✓ · SD ratio 30/25 = **1.2** < 2 ✓

`E` s_p=27.61, SE=6.903, **t=5.65, df=62, p<0.0001**. Since p < α=0.01 we **reject H_0**. There is sufficient evidence that mean ascorbic acid is **higher** for the new method.

No Brown-Forsythe and no quantile plot exist on this route — part d is argued from the text and the SD ratio.

WORKED EXAMPLE 2 · SPORTS ATTENDANCE — matched pairs

Given

`DATA` 5 NBA teams, attendance in 2020 and 2022 — **same teams** → paired · α=0.05
`PATH` Analyze › Specialized Modeling › Matched Pairs › both columns in Y

Answer

`H` μ_d = true mean difference (2022 − 2020). H_0: $\mu_d = 0$ H_1: $\mu_d \neq 0$

`COND` teams **randomly sampled** and independent of each other ✓ · the **differences** show no outliers or skew ✓

`TEST` $\bar{d}$=176,192.8, s_d=34,838.6, **t=11.31, df=4, p=0.0003**. Since p < 0.05 we **reject H_0**. There is sufficient evidence of a difference in mean attendance; 2022 was higher.

`CI` 95% CI (132,935 , 219,451) — **excludes 0**, agreeing with the rejection.

df = **4** (5 pairs − 1), not 8. Run wrongly as independent samples t drops to 5.53.

JMP TEST 2 · BACK

ONE-WAY ANOVA · TWO-WAY ANOVA · TUKEY | ▼ RED TRIANGLE · **GREEN = WRITE IT · FAINT WORD IN A BLANK = WHAT GOES THERE**

T3 · ONE-WAY ANOVA — one factor, 3+ groups

a Run it

`PATH` Analyze › Fit Y by X › Y = response, X = factor › OK
▼ Means/Anova · Unequal Variances · Normal Quantile Plot

X must be **Nominal** (red bars). If it's Continuous you get a regression, not an ANOVA.

b Hypotheses

`SAY` H_0: $\mu_1 = \mu_2 = \mu_3 = \ldots$ (all group means are equal)

`SAY` H_1: **at least one** mean differs from the others

Never write "all means are different". That is wrong and gets marked.

c DF check before you read anything

`CHECK` Factor DF = k−1 · Error DF = N−k · C. Total = N−1

5 groups of 25: 4, 120, 124. Wrong DF → wrong column selected or wrong modeling type.

d Conditions

`SAY` **Independence:** subjects were randomly assigned to the groups.

`SAY` **Normality:** n = *group size* per group and the quantile plots show points near their lines.

`SAY` **Equal variances:** largest SD / smallest SD = *ratio* , under 2, and Brown-Forsythe p = *p-value* > 0.05.

With 3+ groups the rule is **largest over smallest**, not pairwise.

e The F test and your conclusion

`READ` Analysis of Variance table → **F Ratio** and **Prob > F**

`SAY` Since p = *Prob > F* < α = *.05* , we reject H_0. There is sufficient evidence that at least one mean *what + units* differs.

`SAY` Since p = *Prob > F* > α = *.05* , we fail to reject H_0. There is not sufficient evidence that the means differ, so we do **not** make multiple comparisons.

f Multiple comparisons — only if F was significant

▼ Compare Means › **All Pairs, Tukey HSD**

"If appropriate" means exactly this: F not significant → stop. 10 t-tests at 5% each = ~40% false-positive risk; Tukey holds the **overall** rate at 5%.

g Reading connecting letters

`RULE` share a letter → **NOT** different · no letter in common → **different**

`SAY` Tukey HSD comparisons at an overall 0.05 level showed that *groups sharing a letter* did not differ, while *group* was significantly *higher/lower* than *group* .

T4 · TWO-WAY ANOVA — two factors, Fit Model

a Name the factors, their levels, and the response

`SAY` The two factors are *factor A* , **categorical** with *#* levels (*list them*), and *factor B* , **categorical** with *#* levels (*list them*).

`SAY` The response variable is *what + units* , which is **quantitative**.

`SAY` There are *a×b* treatment combinations with *n per cell* units each, so the design is balanced and the interaction can be tested.

b Run it — the Cross step is everything

1. Analyze › **Fit Model** › Y, Response = the response
2. Add **both** factors
3. select **BOTH** inside Construct Model Effects › click **Cross**
4. Run

Three lines must show before Run: *A*, *B*, *A*B*. Only two → you skipped Cross and the interaction is missing. (*Macros › Full Factorial* does all of it in one click.)

c DF check — your safety net

`FORMULA` Model DF = (a−1) + (b−1) + (a−1)(b−1)
3×3, N=4 Model **8** · Error **27** · C. Total **35**
3×2, N=5 Model **5** · Error **24** · C. Total **29**

Model DF too small → no interaction in the model. Go back and Cross. This check catches the wrong-file mistake too.

d Three sets of hypotheses — write all three

`SAY` **Interaction** — H_0: there is no interaction between *factor A* and *factor B* . H_1: there is an interaction.

`SAY` **Factor A** — H_0: $\mu_1 = \mu_2 = \mu_3$. H_1: at least one differs.

`SAY` **Factor B** — H_0: $\mu_1 = \mu_2$. H_1: they differ.

e Read the INTERACTION row first. Always.

`READ` **Effect Tests** table → the **A*B** row → Prob > F

A*B P-VALUE	WHAT YOU MUST DO
< 0.05	**Significant.** Main effects **cannot** be read separately. Compare all **cells**. Draw the circle diagram.
> 0.05	**Not significant.** Say so, then read the two main effect rows separately.

This single decision controls the whole rest of the problem.

f If the interaction IS significant

▼ ON **A*B LSMeans Differences Tukey HSD** → all cell combinations
▼ Factor Profiling › **Interaction Plots** → non-parallel lines confirm it

`SAY` Because the interaction is significant, the main effects cannot be interpreted separately. The effect of _____ on _____ **depends on** _____ .

g If the interaction is NOT significant

▼ ON A FACTOR **LSMeans Differences Tukey HSD** → that factor's levels

Only run Tukey on a **significant** main effect with **3+ levels**. A 2-level factor needs no Tukey — the F test already answered it; just report which level had the higher LSMean.

h The circle diagram, step by step

1. list every cell mean, **sorted high to low**
2. get the cutoff: HSD = q √(MSE/n) — JMP prints it
3. circle each run of means that stays **within** the cutoff
4. circles **overlap** — a mean can sit in two

Same as connecting letters: share a circle → not significantly different. The overlap is the point.

i Summary template

`SAY` A two-way ANOVA examined the effects of _____ and _____ on _____ . The interaction was **[significant / not significant]** (F = _____ , p = _____).

`SAY` *If significant:* the main effects cannot be interpreted separately; the effect of one factor depends on the other. Tukey comparisons of all cells showed _____ .

`SAY` *If not:* the main effects were interpreted separately. _____ was significant (F = _____ , p = _____) and Tukey showed _____ .

j Conditions for ANOVA models

▼ Row Diagnostics › Plot Residual by Predicted → shapeless cloud, no funnel
▼ Save Columns › Residuals → Distribution › Normal Quantile Plot

`SAY` **Independence:** the _____ units were randomly assigned, _____ to each of the _____ treatment combinations. **Normality:** the residual quantile plot shows points near the line. **Equal variances:** the residual by predicted plot shows even spread with no funnel.

Two-way checks **residuals**, not raw data. Cell SD ratio over 2 at n=4–5? Say the estimates are imprecise and lean on the residual plot.

WORKED EXAMPLE 3 · FRUIT FLIES — one-way ANOVA

Given & answer

`DATA` 125 males → **5** groups of 25 · 1 factor, 5 levels · α=0.05
`PATH` Fit Y by X › ▼ Means/Anova › then Compare Means › All Pairs, Tukey HSD

`A` H_0: $\mu_1=\mu_2=\mu_3=\mu_4=\mu_5$ H_1: at least one differs. DF check **4 / 120 / 124**. F is large and p < 0.0001, so **reject H_0** — there is sufficient evidence that reproductive activity affects mean longevity.

`B` Because F was significant, Tukey HSD comparisons were made at an overall 0.05 level. Groups sharing a letter did not differ; the males given **8 virgin females** lived significantly shorter than the others, so reproductive activity **reduces** longevity.

Handout says "four treatments" but the factor has **5** levels.

WORKED EXAMPLE 4 · BATTERY LIFE — two-way ANOVA

Given & answer

`DATA` 3 materials × 3 temps × 4 reps = **36** rows

`A` Factors: **Material Type**, categorical, **3** levels (A,B,C) and **Temperature**, categorical, **3** levels (15, 70, 125°F). Response: **battery life in hours**, quantitative. 9 combinations × 4 each, so the design is balanced and the interaction can be tested.

`DF` Model **8** · Error **27** · C. Total **35**

`EFFECT TESTS` Material F=7.91 p=.0020 · Temp F=28.97 p<.0001 · **Mat*Temp F=3.56 p=.0186**

`B` The **interaction is significant** (p=0.0186), so the main effects cannot be interpreted separately — the effect of temperature **depends on** the material. Tukey on all 9 cells (MSE=675.21, **HSD=61.82**) gives:

`CIRCLES` **A:** B15 155.8, C70 145.8, C15 144.0, A15 134.8, B70 119.8 **B:** C70…B70 + C125 85.5 **C:** C125 85.5, A125 57.5, A70 57.3, B125 49.5

`D` All materials peak at 15°F and fall by 125°F, but A collapses fastest while **C holds up best** as temperature rises. **Recommend material C.**

C125 sits in two circles: matches B70 above and A125 below, yet those two differ from each other.

WHERE EVERY NUMBER LIVES

YOU NEED	WHERE
group SDs	Tests that the Variances are Equal
equal-var p	Brown-Forsythe row
t, df, p pooled	**Pooled t Test** box
t, df, p Welch	**t Test** box
CI for difference	Upper / Lower CL Dif
paired t, df, p	t-Ratio, DF, Prob > |t|
paired CI	Upper95% / Lower95%
one-way F, p	Analysis of Variance table
one-way pairwise	Compare Means › All Pairs, Tukey HSD
interaction test	**Effect Tests**, the A*B row
two-way pairwise	▼ on the effect › LSMeans Differences Tukey HSD
residual check	Row Diagnostics › Plot Residual by Predicted

FIVE TRAPS

Check these before you hand it in

1 "higher / greater / more" = one-sided → Prob > t or Prob < t; "a difference / differ" = two-sided → Prob > |t|. The right one-sided p is under 0.5.

2 Forgetting **Cross** → three effects in the box and Model DF matches the formula, or the interaction never got tested.

3 Reading main effects over a significant interaction → interaction row first, every time.

4 Wrong layout: Fit Y by X = stacked, Matched Pairs = side by side. X must be Nominal.

5 H_1 for ANOVA is **at least one** mean differs, never "all means are different".